JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Matrices and Systems of Linear Equations (2022–2026)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
49	83 M	147 Min	MCQ • MSQ • NAT (2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	28	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	15	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q59 • ID: JAM_2026_Q59) NAT +2 M Neg: Nil (0)

Q.59	Let A be a 3×3 real matrix such that given any column vector $x \in \mathbb{R}^3$, the column vector Ax is the reflection of x about the plane $$\{(a, b, -a-b) : a, b \in \mathbb{R}\}.$$ Then the sum of the diagonal elements of A is ______(*rounded off to one decimal place*).

Question 2 (JAM 2026 • Q57 • ID: JAM_2026_Q57) NAT +2 M Neg: Nil (0)

Q.57	Let $a, b \in \mathbb{R}$. If the system of linear equations $$\begin{aligned} x + 2y + 2z &= 1 \\ 2x + 3y + z &= 2 \\ ax + 5y + bz &= b \end{aligned}$$ has infinitely many solutions, then $a + b =$ ______ (*rounded off to one decimal place*).

Question 3 (JAM 2026 • Q56 • ID: JAM_2026_Q56) NAT +2 M Neg: Nil (0)

Q.56	Let $M \in M_3(\mathbb{R})$. If $$M\begin{pmatrix} \sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ \sqrt{2} \\ -5 \end{pmatrix}, \quad M\begin{pmatrix} 0 \\ 0 \\ \sqrt{30} \end{pmatrix} = \begin{pmatrix} \sqrt{15} \\ \sqrt{10} \\ \sqrt{5} \end{pmatrix}, \quad M\begin{pmatrix} -\sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{12} \\ -\sqrt{18} \\ 0 \end{pmatrix}$$ then $\det(M) =$ ______(*rounded off to one decimal place*).

Question 4 (JAM 2026 • Q50 • ID: JAM_2026_Q50) NAT +1 M Neg: Nil (0)

Q.50	Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P. Then $\det(Q) =$ ________ (*rounded off to one decimal place*).

Question 5 (JAM 2026 • Q49 • ID: JAM_2026_Q49) NAT +1 M Neg: Nil (0)

Q.49	Let $A \in M_3(\mathbb{C})$. Suppose the column vector $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix}$ in $\mathbb{C}^3$ belongs to the intersection of nullspace(A) and rangespace(A^T). Then $\lvert x \rvert =$ ______(*rounded off to one decimal place*).

Question 6 (JAM 2026 • Q47 • ID: JAM_2026_Q47) NAT +1 M Neg: Nil (0)

Q.47	Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30}, then the total number of such distinct surjective homomorphisms is ______.

Question 7 (JAM 2026 • Q39 • ID: JAM_2026_Q39) MSQ +2 M Neg: Nil (0)

Q.39	Consider matrices P of order 4×6 and Q of order 6×4 with real entries such that $PQ = 0$ and $QP = 0$. Which of the following statements is/are TRUE?
(A)	$\text{rangespace}(P) \subseteq \text{nullspace}(Q)$ and $\text{rangespace}(Q) \subseteq \text{nullspace}(P)$.
(B)	$\text{rank}(P) + \text{rank}(Q) \leq 4$.
(C)	If $\text{rangespace}(P) = \text{nullspace}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4$.
(D)	$\text{rangespace}(Q) = \text{nullspace}(P)$.

Question 8 (JAM 2026 • Q37 • ID: JAM_2026_Q37) MSQ +2 M Neg: Nil (0)

Q.37	Let $P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in M_5(\mathbb{C})$. Which of the following statements is/are TRUE?
(A)	$\text{nullity}(P - I) \geq 2$, where I is the 5×5 identity matrix.
(B)	P has 4 distinct eigenvalues in $\mathbb{C}$.
(C)	P has 3 distinct eigenvalues in $\mathbb{R}$.
(D)	If λ is an eigenvalue of P, then there exists a positive integer n such that $\lambda^n = 1$.

Question 9 (JAM 2026 • Q30 • ID: JAM_2026_Q30) MCQ +2 M Neg: −0.67

Q.30	Let G be the power set of $\{1,2,3,4\}$. Let the binary operation Δ on G be $$\mathrm{A}\,\Delta\, B = (A \setminus B) \cup (B \setminus A).$$ Which ONE of the following statements is TRUE?
(A)	$\{1\}$ is an identity element of (G, Δ).
(B)	(G, Δ) is an abelian group but not cyclic.
(C)	(G, Δ) is a group and has an element of order 4.
(D)	(G, Δ) is a group and has an element of order 8.

Question 10 (JAM 2026 • Q25 • ID: JAM_2026_Q25) MCQ +2 M Neg: −0.67

Q.25	Let P be a nonzero complex $n \times n$ matrix such that $v^*Pv \geq 0$ for all column vectors v in $\mathbb{C}^n$. Which ONE of the following statements is necessarily TRUE?
(A)	All eigenvalues of P are negative real numbers.
(B)	If v_1, v_2 are column vectors in $\mathbb{C}^n$ such that $Pv_1 = v_2$ and $Pv_2 = v_1$, then $v_1 = v_2$.
(C)	$v^*Pv \neq 0$ for all nonzero column vectors v in $\mathbb{C}^n$.
(D)	All eigenvalues of P^* are negative real numbers.

Question 11 (JAM 2026 • Q14 • ID: JAM_2026_Q14) **MCQ** +2 M **Neg: −0.67**

Q.14	Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\begin{pmatrix}4\\5\\6\end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?
(A)	$\begin{pmatrix}3\\3\\3\end{pmatrix}$
(B)	$\begin{pmatrix}5\\7\\9\end{pmatrix}$
(C)	$\begin{pmatrix}5\\6\\9\end{pmatrix}$
(D)	$\begin{pmatrix}7\\8\\9\end{pmatrix}$

Question 12 (JAM 2026 • Q8 • ID: JAM_2026_Q8) MCQ +1 M Neg: −0.33

Q.8	Let A be an invertible 5×5 real matrix. Let B be the matrix obtained by interchanging the second and third columns of A and then adding 3 times the third column to the fifth column. Which ONE of the following statements is TRUE?
(A)	B^{-1} is obtained by interchanging the second and third rows of A^{-1}, and then adding 3 times the third row to the fifth row.
(B)	B^{-1} is obtained by interchanging the second and third rows of A^{-1}, and then adding -3 times the fifth row to the third row.
(C)	B^{-1} is obtained by adding -3 times the third row to the fifth row, and then interchanging the second and third rows of A^{-1}.
(D)	B^{-1} is obtained by adding 3 times the fifth row to the third row, and then interchanging the second and third rows of A^{-1}.

Question 13 (JAM 2025 • Q60 • ID: JAM_2025_Q60) NAT +2 M Neg: Nil (0)

Q.60 The number of surjective (onto) group homomorphisms from S_4 to $\mathbb{Z}_6$ is equal to ______________.

Question 14 (JAM 2025 • Q59 • ID: JAM_2025_Q59) NAT +2 M Neg: Nil (0)

Q.59 Let G be an abelian group of order 35. Let m denote the number of elements of order 5 in G, and let n denote the number of elements of order 7 in G.

Then, the value of $m + n$ is equal to ______________.

Question 15 (JAM 2025 • Q52 • ID: JAM_2025_Q52)   NAT   +2 M   Neg: Nil (0)

Q.52 Let $\sigma \in S_4$ be the permutation defined by $\sigma(1) = 2,\ \sigma(2) = 3,\ \sigma(3) = 1$ and $\sigma(4) = 4$. The number of elements in the set

$$\{\tau \in S_4 : \tau\sigma\tau^{-1} = \sigma\}$$

is equal to ____________.

Question 16 (JAM 2025 • Q39 • ID: JAM_2025_Q39)   MSQ   +2 M   Neg: Nil (0)

Q.39 For $n \in \mathbb{N}$, consider the set $U(n) = \{\bar{x} \in \mathbb{Z}_n : \gcd(x, n) = 1\}$ as a group under multiplication modulo n.

Then, which of the following is/are TRUE?

(A) $U(8)$ is a cyclic group

(B) $U(5)$ is a cyclic group

(C) $U(12)$ is a cyclic group

(D) $U(9)$ is a cyclic group

Question 17 (JAM 2025 • Q23 • ID: JAM_2025_Q23) MCQ +2 M Neg: −0.67

Q.23 Let $\mathbb{R}/\mathbb{Z}$ denote the quotient group, where $\mathbb{Z}$ is considered as a subgroup of the additive group of real numbers $\mathbb{R}$.

Let m denote the number of injective (one-one) group homomorphisms from $\mathbb{Z}_3$ to $\mathbb{R}/\mathbb{Z}$ and n denote the number of group homomorphisms from $\mathbb{R}/\mathbb{Z}$ to $\mathbb{Z}_3$.

Then, which one of the following is TRUE?

(A) $m = 2$ and $n = 1$

(B) $m = 3$ and $n = 3$

(C) $m = 2$ and $n = 3$

(D) $m = 1$ and $n = 1$

Question 18 (JAM 2025 • Q22 • ID: JAM_2025_Q22) MCQ +2 M Neg: −0.67

Q.22 Let $M = (m_{ij})$ be a 3×3 real, invertible matrix and $\sigma \in S_3$ be the permutation defined by $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$. The matrix $M_\sigma = (n_{ij})$ is defined by $n_{ij} = m_{i\sigma(j)}$ for all $i, j \in \{1, 2, 3\}$.

Then, which one of the following is TRUE?

(A) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

(B) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1

(C) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1

(D) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

Question 19 (JAM 2025 • Q16 • ID: JAM_2025_Q16) MCQ +2 M Neg: −0.67

Q.16 Let $M = \begin{pmatrix} -2 & 0 & 0 \\ 3 & 2 & 3 \\ 4 & -1 & x \end{pmatrix}$ for some real number x. Suppose that -2 and 3 are eigenvalues of M. If $M^3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 125 \\ 125 \end{pmatrix}$, then which one of the following is TRUE?

(A) $x = 5$, and the matrix $M^2 + M$ is invertible

(B) $x \neq 5$, and the matrix $M^2 + M$ is invertible

(C) $x = 5$, and the matrix $M^2 + M$ is NOT invertible

(D) $x \neq 5$, and the matrix $M^2 + M$ is NOT invertible

Question 20 (JAM 2025 • Q15 • ID: JAM_2025_Q15) MCQ +2 M Neg: −0.67

Q.15 Let $M = \begin{pmatrix} 6 & 2 & -6 & 8 \\ 5 & 3 & -9 & 8 \\ 3 & 1 & -2 & 4 \end{pmatrix}$. Consider the system S of linear equations given by

$$\begin{aligned} 6x_1 + 2x_2 - 6x_3 + 8x_4 &= 8 \\ 5x_1 + 3x_2 - 9x_3 + 8x_4 &= 16 \\ 3x_1 + x_2 - 2x_3 + 4x_4 &= 32 \end{aligned}$$

where x_1, x_2, x_3, x_4 are unknowns.

Then, which one of the following is TRUE?

(A) The rank of M is 3, and the system S has a solution

(B) The rank of M is 3, and the system S does NOT have a solution

(C) The rank of M is 2, and the system S has a solution

(D) The rank of M is 2, and the system S does NOT have a solution

Question 21 (JAM 2025 • Q13 • ID: JAM_2025_Q13) **MCQ** **+2 M** **Neg: −0.67**

Q.13 Let $X = \{x \in S_4 : x^3 = id\}$ and $Y = \{x \in S_4 : x^2 \neq id\}$.

If m and n denote the number of elements in X and Y, respectively, then which one of the following is TRUE?

(A) m is even and n is even

(B) m is odd and n is even

(C) m is even and n is odd

(D) m is odd and n is odd

Question 22 (JAM 2025 • Q9 • ID: JAM_2025_Q9) **MCQ** **+1 M** **Neg: −0.33**

Q.9 Let G be a finite abelian group of order 10. Let x_0 be an element of order 2 in G.

If $X = \{x \in G : x^3 = x_0\}$, then which one of the following is TRUE?

(A) X has exactly one element

(B) X has exactly two elements

(C) X has exactly three elements

(D) X is an empty set

Question 23 (JAM 2025 • Q7 • ID: JAM_2025_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $M = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & x \end{pmatrix}$ for some real number x.

If 0 is an eigenvalue of M, then $(M^4 + M)\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is equal to

(A) $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$

(D) $\begin{pmatrix} 17 \\ 0 \\ 17 \end{pmatrix}$

Question 24 (JAM 2024 • Q50 • ID: JAM_2024_Q50) NAT +1 M Neg: Nil (0)

Q.50 Let

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}.$$

If $p(x)$ is the characteristic polynomial of M, then $p(2) - 1$ equals __________

Question 25 (JAM 2024 • Q44 • ID: JAM_2024_Q44) NAT +1 M Neg: Nil (0)

Q.44 Consider the 4×4 matrix

$$M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}.$$

If $a_{i,j}$ denotes the $(i,j)^{\text{th}}$ entry of M^{-1}, then $a_{4,1}$ equals __________ (*rounded off to two decimal places*).

Question 26 (JAM 2024 • Q37 • ID: JAM_2024_Q37) MSQ +2 M Neg: Nil (0)

Q.37 For a matrix M, let Rowspace(M) denote the linear span of the rows of M and Colspace(M) denote the linear span of the columns of M. Which of the following hold(s) for all $A, B, C \in M_{10}(\mathbb{R})$ satisfying $A = BC$?

(A) Rowspace$(A) \subseteq$ Rowspace(B)

(B) Rowspace$(A) \subseteq$ Rowspace(C)

(C) Colspace$(A) \subseteq$ Colspace(B)

(D) Colspace$(A) \subseteq$ Colspace(C)

Question 27 (JAM 2024 • Q16 • ID: JAM_2024_Q16) **MCQ** **+2 M** **Neg: −0.67**

Q.16 Let A be a 6×5 matrix with entries in $\mathbb{R}$ and B be a 5×4 matrix with entries in $\mathbb{R}$. Consider the following two statements.

P: For all such nonzero matrices A and B, there is a nonzero matrix Z such that AZB is the 6×4 zero matrix.

Q: For all such nonzero matrices A and B, there is a nonzero matrix Y such that BYA is the 5×5 zero matrix.

Which one of the following holds?

(A) Both P and Q are true

(B) P is true but Q is false

(C) P is false but Q is true

(D) Both P and Q are false

Question 28 (JAM 2024 • Q15 • ID: JAM_2024_Q15) MCQ +2 M Neg: −0.67

Q.15

Let $a = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \\ 0 \end{bmatrix}$. Consider the following two statements.

P: The matrix $I_4 - aa^T$ is invertible.

Q: The matrix $I_4 - 2aa^T$ is invertible.

Then, which one of the following holds?

(A) P is false but Q is true

(B) P is true but Q is false

(C) Both P and Q are true

(D) Both P and Q are false

Question 29 (JAM 2024 • Q9 • ID: JAM_2024_Q9) **MCQ** +1 M **Neg: −0.33**

Q.9 Consider the following statements.

P: If a system of linear equations $Ax = b$ has a unique solution, where A is an $m \times n$ matrix and b is an $m \times 1$ matrix, then $m = n$.

Q: For a subspace W of a nonzero vector space V, whenever $u \in V \setminus W$ and $v \in V \setminus W$, then $u + v \in V \setminus W$.

Which one of the following holds?

(A) Both P and Q are true

(B) P is true but Q is false

(C) P is false but Q is true

(D) Both P and Q are false

Question 30 (JAM 2024 • Q6 • ID: JAM_2024_Q6) MCQ +1 M Neg: −0.33

Q.6 For a positive integer n, let $U(n) = \{\bar{r} \in \mathbb{Z}_n : \gcd(r, n) = 1\}$ be the group under multiplication modulo n. Then, which one of the following statements is TRUE?

(A) $U(5)$ is isomorphic to $U(8)$

(B) $U(10)$ is isomorphic to $U(12)$

(C) $U(8)$ is isomorphic to $U(10)$

(D) $U(8)$ is isomorphic to $U(12)$

Question 31 (JAM 2024 • Q5 • ID: JAM_2024_Q5) MCQ +1 M Neg: −0.33

Q.5 Let G be a group of order 39 such that it has exactly one subgroup of order 3 and exactly one subgroup of order 13. Then, which one of the following statements is TRUE?

(A) G is necessarily cyclic

(B) G is abelian but need not be cyclic

(C) G need not be abelian

(D) G has 13 elements of order 13

Question 32 (JAM 2023 • Q58 • ID: JAM_2023_Q58) **NAT** **+2 M** **Neg: Nil (0)**

Q. 58 The maximum number of linearly independent eigenvectors of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is equal to ____________.

Question 33 (JAM 2023 • Q56 • ID: JAM_2023_Q56) **NAT** **+2 M** **Neg: Nil (0)**

Q. 56 Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the maximum possible rank of B is equal to ____________.

Question 34 (JAM 2023 • Q14 • ID: JAM_2023_Q14) MCQ +2 M Neg: −0.67

Q. 14 The system of linear equations in x_1, x_2, x_3

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{R}$, has

(A) at least one solution for any α and β

(B) a unique solution for any β when $\alpha \neq 1$

(C) NO solution for any α when $\beta \neq 5$

(D) infinitely many solutions for any α when $\beta = 5$

Question 35 (JAM 2023 • Q13 • ID: JAM_2023_Q13) MCQ +2 M Neg: −0.67

Q. 13 Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$$

and $B = A^5 + A^4 + I_3$. Which of the following is NOT an eigenvalue of B?

(A) 1

(B) 2

(C) 49

(D) 3

Question 36 (JAM 2023 • Q2 • ID: JAM_2023_Q2) MCQ +1 M Neg: −0.33

Q. 2 Let $\mathbf{v}_1, \ldots, \mathbf{v}_9$ be the column vectors of a non-zero 9×9 real matrix A. Let $a_1, \ldots, a_9 \in \mathbb{R}$, not all zero, be such that $\sum_{i=1}^{9} a_i \mathbf{v}_i = \mathbf{0}$. Then the system $A\mathbf{x} = \sum_{i=1}^{9} \mathbf{v}_i$ has

(A) no solution

(B) a unique solution

(C) more than one but only finitely many solutions

(D) infinitely many solutions

Question 37 (JAM 2022 • Q57 • ID: JAM_2022_Q57) NAT +2 M Neg: Nil (0)

Q.57 Consider the 4×4 matrix $M = \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix}$. Then the value of the determinant of M is equal to ________.

Question 38 (JAM 2022 • Q53 • ID: JAM_2022_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$ and let A^{T} denote the transpose of A. Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be column vectors with entries in $\mathbb{R}$ such that $u_1^2 + u_2^2 = 1$ and $v_1^2 + v_2^2 + v_3^2 = 1$. Suppose

$$Au = \sqrt{2}\, v \quad \text{and} \quad A^{\mathsf{T}} v = \sqrt{2}\, u.$$

Then $|u_1 + 2\sqrt{2}\, v_1|$ is equal to ________. (*Rounded off to two decimal places*)

Question 39 (JAM 2022 • Q40 • ID: JAM_2022_Q40) MSQ +2 M Neg: Nil (0)

Q.40	Let P be a fixed 3×3 matrix with entries in $\mathbb{R}$. Which of the following maps from $M_3(\mathbb{R})$ to $M_3(\mathbb{R})$ is/are linear?
(A)	$T_1 : M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_1(M) = MP - PM$ for $M \in M_3(\mathbb{R})$.
(B)	$T_2 : M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_2(M) = M^2P - P^2M$ for $M \in M_3(\mathbb{R})$.
(C)	$T_3 : M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_3(M) = MP^2 + P^2M$ for $M \in M_3(\mathbb{R})$.
(D)	$T_4 : M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_4(M) = MP^2 - PM^2$ for $M \in M_3(\mathbb{R})$.

Section C: Q.41 – Q.50 Carry ONE mark each.

Question 40 (JAM 2022 • Q39 • ID: JAM_2022_Q39) MSQ +2 M Neg: Nil (0)

Q.39	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 5, together with the zero polynomial. Let $T: V \to \mathbb{R}$ be the linear map defined by $T(1) = 1$ and $$T\big(x(x-1)\cdots(x-k+1)\big) = 1 \quad \text{for } 1 \le k \le 5.$$ Then which of the following is/are true?
(A)	$T(x^4) = 15.$
(B)	$T(x^3) = 5.$
(C)	$T(x^4) = 14.$
(D)	$T(x^3) = 3.$

Question 41 (JAM 2022 • Q30 • ID: JAM_2022_Q30) MCQ +2 M Neg: −0.67

Q.30	Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3, respectively. Then the entry in the first row and the third column of P is
(A)	0
(B)	1
(C)	−1
(D)	2

Question 42 (JAM 2022 • Q26 • ID: JAM_2022_Q26) **MCQ** **+2 M** **Neg: −0.67**

Q.26	For a 4×4 matrix $M \in M_4(\mathbb{C})$, let $\overline{M}$ denote the matrix obtained from M by replacing each entry of M by its complex conjugate. Consider the real vector space $$H = \{M \in M_4(\mathbb{C}) : \quad M^{\mathsf{T}} = \overline{M}\}$$ where M^{T} denotes the transpose of M. The dimension of H as a vector space over $\mathbb{R}$ is equal to
(A)	6
(B)	16
(C)	15
(D)	12

Question 43 (JAM 2022 • Q20 • ID: JAM_2022_Q20) MCQ +2 M Neg: −0.67

Q.20	For $P \in M_5(\mathbb{R})$ and $i, j \in \{1,2,\ldots,5\}$, let p_{ij} denote the $(i,j)^{\text{th}}$ entry of P. Let $$S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs} \text{ for } i,j,r,s \in \{1,2,\ldots,5\} \text{ with } i + r = j + s\}.$$ Then which one of the following is FALSE?
(A)	S is a subspace of the vector space over $\mathbb{R}$ of all 5×5 symmetric matrices.
(B)	The dimension of S over $\mathbb{R}$ is 5.
(C)	The dimension of S over $\mathbb{R}$ is 11.
(D)	If $P \in S$ and all the entries of P are integers, then 5 divides the sum of all the diagonal entries of P.

Question 44 (JAM 2022 • Q18 • ID: JAM_2022_Q18) MCQ +2 M Neg: −0.67

Q.18	Consider the system of linear equations $$\begin{aligned} x+y+t &= 4, \\ 2x-4t &= 7, \\ x+y+z &= 5, \\ x-3y-z-10t &= \lambda, \end{aligned}$$ where x, y, z, t are variables and λ is a constant. Then which one of the following is true?
(A)	If $\lambda = 1$, then the system has a unique solution.
(B)	If $\lambda = 2$, then the system has infinitely many solutions.
(C)	If $\lambda = 1$, then the system has infinitely many solutions.
(D)	If $\lambda = 2$, then the system has a unique solution.

Question 45 (JAM 2022 • Q17 • ID: JAM_2022_Q17) MCQ +2 M Neg: −0.67

Q.17	For $X, Y \in M_2(\mathbb{R})$, define $(X, Y) = XY - YX$. Let $\mathbf{0} \in M_2(\mathbb{R})$ denote the zero matrix. Consider the two statements: P : $\big(X,(Y,Z)\big) + \big(Y,(Z,X)\big) + \big(Z,(X,Y)\big) = \mathbf{0}$ for all $X, Y, Z \in M_2(\mathbb{R})$. Q : $\big(X,(Y,Z)\big) = \big((X,Y),Z\big)$ for all $X, Y, Z \in M_2(\mathbb{R})$. Then which one of the following is correct?
(A)	Both P and Q are true.
(B)	P is true, but Q is false.
(C)	P is false, but Q is true.
(D)	Both P and Q are false.

Question 46 (JAM 2022 • Q16 • ID: JAM_2022_Q16) MCQ +2 M Neg: −0.67

Q.16	Let $P \in M_4(\mathbb{R})$ be such that P^4 is the zero matrix, but P^3 is a nonzero matrix. Then which one of the following is FALSE?
(A)	For every nonzero vector $v \in \mathbb{R}^4$, the subset $\{v, Pv, P^2v, P^3v\}$ of the real vector space $\mathbb{R}^4$ is linearly independent.
(B)	The rank of P^k is $4 - k$ for every $k \in \{1,2,3,4\}$.
(C)	0 is an eigenvalue of P.
(D)	If $Q \in M_4(\mathbb{R})$ is such that Q^4 is the zero matrix, but Q^3 is a nonzero matrix, then there exists a nonsingular matrix $S \in M_4(\mathbb{R})$ such that $S^{-1}QS = P$.

Question 47 (JAM 2022 • Q3 • ID: JAM_2022_Q3) MCQ +1 M Neg: −0.33

Q.3	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 6, together with the zero polynomial. Then which one of the following is true?
(A)	$\{f \in V : f(1/2) \notin \mathbb{Q}\}$ is a subspace of V.
(B)	$\{f \in V : f(1/2) = 1\}$ is a subspace of V.
(C)	$\{f \in V : f(1/2) = f(1)\}$ is a subspace of V.
(D)	$\{f \in V : f'(1/2) = 1\}$ is a subspace of V.

Question 48 (JAM 2022 • Q2 • ID: JAM_2022_Q2) MCQ +1 M Neg: −0.33

Q.2	The rank of the 4×6 matrix $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ with entries in $\mathbb{R}$, is
(A)	1
(B)	2
(C)	3
(D)	4

Question 49 (JAM 2022 • Q1 • ID: JAM_2022_Q1) MCQ +1 M Neg: −0.33

Q.1	Consider the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$. If the eighth power of M satisfies $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then the value of x is
(A)	21
(B)	22
(C)	34
(D)	35

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Matrices and Systems of Linear Equations (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**1.0 to 1.0**
2	NAT	+2	**6.0 to 6.0**
3	NAT	+2	**−1.0 to −1.0**
4	NAT	+1	**16.0 to 16.0**
5	NAT	+1	**3.0 to 3.0**
6	NAT	+1	**8 to 8**
7	MSQ	+2	**A;B;C**
8	MSQ	+2	**A;B;D**
9	MCQ	+2	**B**
10	MCQ	+2	**B**
11	MCQ	+2	**D**
12	MCQ	+1	**B**
13	NAT	+2	**0 to 0**
14	NAT	+2	**10 to 10**
15	NAT	+2	**3 to 3**
16	MSQ	+2	**B;D**
17	MCQ	+2	**A**
18	MCQ	+2	**C**
19	MCQ	+2	**B**
20	MCQ	+2	**A**
21	MCQ	+2	**B**
22	MCQ	+1	**A**
23	MCQ	+1	**B**
24	NAT	+1	**31 to 31**
25	NAT	+1	**0.15 to 0.18**
26	MSQ	+2	**B;C**
27	MCQ	+2	**A**
28	MCQ	+2	**A**
29	MCQ	+1	**D**
30	MCQ	+1	**D**
31	MCQ	+1	**A**
32	NAT	+2	**3 to 3**
33	NAT	+2	**2 to 2**
34	MCQ	+2	**B**
35	MCQ	+2	**B**
36	MCQ	+1	**D**
37	NAT	+2	**41 to 41**
38	NAT	+2	**3.00 to 3.00**
39	MSQ	+2	**A,C**
40	MSQ	+2	**A,B**
41	MCQ	+2	**C**
42	MCQ	+2	**B**
43	MCQ	+2	**C**
44	MCQ	+2	**C**
45	MCQ	+2	**B**
46	MCQ	+2	**A**
47	MCQ	+1	**C**
48	MCQ	+1	**D**
49	MCQ	+1	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.